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United States Patent	12399237
Kind Code	B2
Date of Patent	August 26, 2025
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Magnetic sensor and magnetic detection system

Abstract

A magnetic sensor includes at least one bias magnet, a first half-bridge circuit, a second half-bridge circuit, and a base member. The first half-bridge circuit includes a pair of first magnetoresistive effect elements to one of which a bias magnetic field aligned with a positive direction of an X-axis is applied and to the other of which a bias magnetic field aligned with a negative direction of the X-axis is applied. The second half-bridge circuit includes a pair of second magnetoresistive effect elements to one of which a bias magnetic field aligned with a positive direction of a Y-axis is applied and to the other of which a bias magnetic field aligned with a negative direction of the Y-axis is applied.

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Appl. No.:	18/561565
Filed (or PCT Filed):	May 16, 2022
PCT No.:	PCT/JP2022/020401
PCT Pub. No.:	WO2022/244734
PCT Pub. Date:	November 24, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240248157 A1	Jul. 25, 2024

Foreign Application Priority Data

JP 2021-083385 May. 17, 2021

Publication Classification

Int. Cl.: G01R33/09 (20060101); H10N50/85 (20230101); H10N50/10 (20230101)

U.S. Cl.:

CPC G01R33/093 (20130101); H10N50/85 (20230201); H10N50/10 (20230201)

Field of Classification Search

CPC: G01R (33/093)

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Background/Summary

(1) This application is the U.S. National Phase under 35 U.S.C. § 371 of International Patent Application No. PCT/JP2022/020401, filed on May 16, 2022, which in turn claims the benefit of Japanese Patent Application No. 2021-083385, filed on May 17, 2021, the entire disclosures of which Applications are incorporated by reference herein.

TECHNICAL FIELD

(2) The present disclosure generally relates to magnetic sensors and magnetic detection systems and specifically relates to a magnetic sensor, including at least one bias magnet, and a magnetic detection system.

BACKGROUND ART

(3) Patent Literature 1 describes a rotational angle determining sensor (magnetic sensor) configured as explained below. At least two pairs of GMR elements each having a free magnetic layer and a pinning magnetic layer are disposed on a substrate. The GMR elements in each pair are connected to each other in series, a rotatably magnet is disposed to face the GMR elements, and this magnet applies a saturated magnetic field to the GMR elements. A change in resistance values of the GMR elements caused due to an angle formed between an orientation of magnetization of the free magnetic layer following an orientation of a magnetic field line of a magnet and a direction of magnetization of the pinning magnetic layer determines a rotational angle of a rotational axis of the magnet.

CITATION LIST

Patent Literature

(4) Patent Literature 1: JP 2002-303536 A

SUMMARY OF INVENTION

(5) An object of the present disclosure is to provide a magnetic sensor and a magnetic detection system, each contributing to more accurately detecting the orientation of a magnetic field applied to a magnetic sensor.

(6) A magnetic sensor according to an aspect of the present disclosure includes at least one bias magnet, a first half-bridge circuit, a second half-bridge circuit, and a base member. The at least one bias magnet is configured to generate a bias magnetic field aligned with a positive direction of an X-axis, a bias magnetic field aligned with a negative direction of the X-axis, a bias magnetic field aligned with a positive direction of a Y-axis orthogonal to the X-axis, and a bias magnetic field aligned with a negative direction of the Y-axis. The base member holds the at least one bias magnet, the first half-bridge circuit, and the second half-bridge circuit. The first half-bridge circuit includes a pair of first magnetoresistive effect elements half-bridge connected to each other and configured to detect a magnetic field aligned with the X-axis and a first output terminal configured to output a first output signal from a connection point between the pair of first magnetoresistive effect elements. The second half-bridge circuit includes a pair of second magnetoresistive effect elements half-bridge connected to each other and configured to detect a magnetic field aligned with the Y-axis and a second output terminal configured to output a second output signal from a connection point between the pair of second magnetoresistive effect elements. The bias magnetic field aligned with the positive direction of the X-axis is applied to one of the pair of first magnetoresistive effect elements, and the bias magnetic field aligned with the negative direction of the X-axis is applied to the other of the pair of first magnetoresistive effect elements. The bias magnetic field aligned with the positive direction of the Y-axis is applied to one of the pair of second magnetoresistive effect elements, and the bias magnetic field aligned with the negative direction of the Y-axis is applied to the other of the pair of second magnetoresistive effect elements.

(7) A magnetic detection system according to another aspect of the present disclosure includes the

magnetic sensor described above and a processing circuit. The processing circuit determines, based on at least the first output signal and the second output signal, an orientation of a magnetic field applied to the magnetic sensor.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a plan view of a magnetic sensor according to an embodiment;
- (2) FIG. 2 is a plan view of the magnetic sensor, in which a bias magnet is omitted;
- (3) FIG. 3 is a sectional view of the magnetic sensor;
- (4) FIG. 4 is a schematic diagram illustrating how the magnetic sensor may be used;
- (5) FIG. 5 is an equivalent circuit diagram of a first half-bridge circuit and a third half-bridge circuit of the magnetic sensor;
- (6) FIG. 6 is an equivalent circuit diagram of a second half-bridge circuit and a fourth half-bridge circuit of the magnetic sensor;
- (7) FIG. 7 is a cross-sectional view of a magnetoresistive effect element of the magnetic sensor; and
- (8) FIG. 8 shows output signals of the magnetic sensor.

DESCRIPTION OF EMBODIMENTS

(9) A magnetic sensor and magnetic detection system according to an embodiment will be described with reference to the accompanying drawings. Note that the embodiment to be described below is only an exemplary one of various embodiments of the present disclosure and should not be construed as limiting. Rather, the embodiment which will be described below may be readily modified in various manners depending on a design choice or any other factor without departing from the scope of the present disclosure. The drawings to be referred to in the following description of embodiments are all schematic representations. Thus, the ratio of the dimensions (including thicknesses) of respective constituent elements illustrated on the drawings does not always reflect their actual dimensional ratio.

(10) (Overview)

(11) As shown in FIG. 1, a magnetic sensor **100** of the present embodiment includes at least one bias magnet **5**, a first half-bridge circuit **1**, a second half-bridge circuit **2**, and a base member **73** (see FIG. 3). The at least one bias magnet **5** generates a bias magnetic field aligned with a positive direction of an X-axis, a bias magnetic field aligned with a negative direction of the X-axis, a bias magnetic field aligned with a positive direction of a Y-axis orthogonal to the X-axis, and a bias magnetic field aligned with a negative direction of the Y-axis. The base member **73** holds the at least one bias magnet **5**, the first half-bridge circuit **1**, and the second half-bridge circuit **2**. The first half-bridge circuit **1** includes: a pair of first magnetoresistive effect elements **1P** and **1Q** half-bridge connected to each other and configured to detect a magnetic field aligned with the X-axis; and a first output terminal **1T** configured to output a first output signal from a connection point between the pair of first magnetoresistive effect elements **1P** and **1Q**. The second half-bridge circuit **2** includes; a pair of second magnetoresistive effect elements **2P** and **2Q** half-bridge connected to each other and configured to detect a magnetic field aligned with the Y-axis; and a second output terminal **2T** configured to output a second output signal from a connection point between the pair of second magnetoresistive effect elements **2P** and **2Q**. The bias magnetic field aligned with the positive direction of the X-axis is applied to one of the pair of first magnetoresistive effect elements **1P** and **1Q**, and the bias magnetic field aligned with the negative direction of the X-axis is applied to the other of the pair of first magnetoresistive effect elements **1P** and **1Q**. The bias magnetic field aligned with the positive direction of the Y-axis is applied to one of the pair of second magnetoresistive effect elements **2P** and **2Q**, and the bias magnetic field aligned with the negative

direction of the Y-axis is applied to the other of the pair of second magnetoresistive effect elements 2P and 2Q.

(12) This embodiment causes the first output signal delivered as the magnetic field applied to the magnetic sensor **100** rotates to have a waveform close to an ideal sinusoidal wave and causes the second output signal to have a waveform close to an ideal cosine wave. This enables accurately determining, based on the first output signal and the second output signal, the orientation of the magnetic field applied to the magnetic sensor **100**.

(13) Moreover, the first half-bridge circuit **1** and the second half-bridge circuit **2** are aggregated together in the base member **73**. This saves the trouble of adjusting the positional relationship between the first base member and the second base member, compared to a situation where a first base member on which the first half-bridge circuit **1** is mounted and a second base member on which the second half-bridge circuit **2** is mounted are individually disposed. In addition, this also reduces the chances of a change in positional relation causing a decline in the accuracy of the orientation of the magnetic field detected.

(14) To say that the direction of the bias magnetic field or the detection direction of a magnetic field by the magnetoresistive effect element is aligned with the X-axis or the Y-axis, the angular difference between the direction and the axis is preferably less than or equal to 5 degrees.

(15) In the following description, a Z-axis, which is an axis perpendicular to both the X-axis and the Y-axis, will be used along with the X-axis and the Y-axis. Note that each of the X-axis, the Y-axis, and the Z-axis is a virtual axis defined on the magnetic sensor **100** and is an insubstantial one.

(16) (Details)

(17) (1) Overall Configuration

(18) As shown in FIGS. **1** to **3**, the magnetic sensor **100** includes a second protection film **72**, the bias magnet **5**, a first protection film **71**, a wiring layer **W1**, and the base member **73**. The wiring layer **W1** includes a first half-bridge circuit **1**, the second half-bridge circuit **2**, a third half-bridge circuit **3**, and a fourth half-bridge circuit **4**. Note that in FIG. **1**, only the wiring layer **W1** and the bias magnet **5** are shown, and in FIG. **2**, only the wiring layer **W1** is shown.

(19) A magnetic detection system **200** includes the magnetic sensor **100** and a processing circuit **201**. The processing circuit **201** determines, based on at least the first output signal and the second output signal, the orientation of a magnetic field applied to the magnetic sensor **100**.

(20) In the following description of the embodiment, the magnetic sensor **100** and the magnetic detection system **200** are supposed to be used to determine the orientation of a magnetic field generated from a rotor **8** (refer to FIG. **4**) of a motor and thereby determine the rotational angle of the rotor **8**.

(21) (2) Rotor

(22) The rotor **8** includes a plurality of permanent magnets. The plurality of permanent magnets form a plurality of magnetic poles **80**. In other words, the plurality of permanent magnets have the plurality of magnetic poles **80**. The plurality of magnetic poles **80** are arranged side by side in the rotational direction of the rotor **8** such that N-poles and S-poles are alternately arranged. In FIG. **4**, the plurality of magnetic poles **80** are arranged side by side such that the N- and S-poles alternate with each other every 45 degrees in the rotational direction of the rotor **8**. In FIG. **4**, each of the magnetic poles **80** is labeled as either “N” indicating that the magnetic pole **80** is an N-pole or “S” indicating that the magnetic pole **80** is an S-pole. Note that the letters “N” and “S” are shown on the drawings for illustrative purposes only and actually the magnetic poles **80** do not have such inscribed letters. In FIGS. **1** and **3**, the same applies to “N” and “S” shown on the bias magnet **5**.

(23) (3) Bias Magnet

(24) As shown in FIGS. **1** and **3**, the bias magnet **5** is in the shape of a rectangular parallelepiped. The bias magnet **5** is a single member. As the bias magnet **5**, for example, a permanent magnet or an electromagnet may be employed. The bias magnet **5** of the present embodiment is a permanent magnet. The bias magnet **5** is, for example, a ferrite magnet or a neodymium magnet.

(25) The bias magnet **5** includes a plurality of (in the present embodiment, eight) magnetic poles **50**. Four magnetic poles **50** of the eight magnetic poles **50** are disposed on a first plane parallel to both the X-axis and the Y-axis. The remaining four magnetic poles **50** of the eight magnetic poles **50** are disposed on a second plane parallel to the first plane.

(26) That is, two groups each including four magnetic poles **50** are disposed. In each group, the four magnetic poles **50** are on the same plane. The magnetic poles **50** belonging to different groups are disposed at different positions in a Z-axis direction. A Z coordinate of each of the four magnetic poles **50** shown in FIG. **1** is greater than a Z coordinate of each of the remaining four magnetic poles **50**.

(27) The eight magnetic poles **50** are disposed such that the magnetic poles **50** adjacent to each other in an X-axis direction have different poles and the magnetic pole **50** adjacent to each other in a Y-axis direction have different poles. Moreover, the eight magnetic poles **50** are disposed such that the magnetic poles **50** adjacent to each other in the Z-axis direction have different poles.

(28) (4) Base Member

(29) As shown in FIG. **3**, the base member **73** has a plate shape. The base member **73** is, for example, an alumina substrate.

(30) (5) Wiring Layer

(31) As shown in FIG. **3**, the wiring layer **W1** is formed on a surface of the base member **73**. Thus, the base member **73** holds the wiring layer **W1**.

(32) The wiring layer **W1** of the present embodiment includes, for example, a plurality of layers. The plurality of layers are electrically connected to each other via a through hole.

(33) As shown in FIG. **2**, the wiring layer **W1** includes the first half-bridge circuit **1**, the second half-bridge circuit **2**, the third half-bridge circuit **3**, and the fourth half-bridge circuit **4**.

(34) The first half-bridge circuit **1** includes the pair of first magnetoresistive effect elements **1P** and **1Q** and the first output terminal **1T**.

(35) The second half-bridge circuit **2** includes the pair of second magnetoresistive effect elements **2P** and **2Q** and the second output terminal **2T**.

(36) As shown in FIG. **5**, the third half-bridge circuit **3** includes a pair of third magnetoresistive effect elements **3P** and **3Q** and a third output terminal **3T**. The pair of third magnetoresistive effect elements **3P** and **3Q** are half-bridge connected to each other. The pair of third magnetoresistive effect elements **3P** and **3Q** detect a magnetic field aligned with the X-axis. The third output terminal **3T** outputs a third output signal from a connection point between the pair of third magnetoresistive effect elements **3P** and **3Q**.

(37) As shown in FIG. **6**, the fourth half-bridge circuit **4** includes a pair of fourth magnetoresistive effect elements **4P** and **4Q** and a fourth output terminal **4T**. The pair of fourth magnetoresistive effect elements **4P** and **4Q** are half-bridge connected to each other. The pair of fourth magnetoresistive effect elements **4P** and **4Q** detect a magnetic field aligned with the Y-axis. The fourth output terminal **4T** outputs a fourth output signal from a connection point between the pair of fourth magnetoresistive effect elements **4P** and **4Q**.

(38) In the following description, each of the first magnetoresistive effect elements **1P** and **1Q**, the second magnetoresistive effect elements **2P** and **2Q**, the third magnetoresistive effect elements **3P** and **3Q**, and the fourth magnetoresistive effect elements **4P** and **4Q** will sometimes be referred to as a magnetoresistive effect element **Mr0**. That is, the magnetic sensor **100** includes a plurality of (eight) magnetoresistive effect elements **Mr0**.

(39) As shown in FIG. **2**, the wiring layer **W1** further includes power supply terminals **H10** and **H20** and reference terminals **L10** and **L20**. The power supply terminals **H10** and **H20** are higher-potential terminals electrically connected to a higher-potential electrical path of a power supply. The reference terminals **L10** and **L20** are lower-potential terminals electrically connected to a lower-potential electrical path (electrical path of a reference potential) of the power supply. In the present embodiment, the reference terminals **L10** and **L20** are grounding terminals electrically

connected an electrical path of a ground potential.

(40) The first magnetoresistive effect element **1P** has a first terminal electrically connected to the reference terminal **L20**. The first magnetoresistive effect element **1P** has a second terminal electrically connected to a first terminal of the first magnetoresistive effect element **1Q**. The first magnetoresistive effect element **1Q** has a second terminal electrically connected to the power supply terminal **H10**. The first output terminal **1T** is electrically connected to the connection point between the pair of first magnetoresistive effect elements **1P** and **1Q**.

(41) The second magnetoresistive effect element **2P** has a first terminal electrically connected to the power supply terminal **H10**. The second magnetoresistive effect element **2P** has a second terminal electrically connected to a first terminal of the second magnetoresistive effect element **2Q**. The second magnetoresistive effect element **2Q** has a second terminal electrically connected to the reference terminal **L10**. The second output terminal **2T** is electrically connected to the connection point between the pair of second magnetoresistive effect elements **2P** and **2Q**.

(42) The third magnetoresistive effect element **3P** has a first terminal electrically connected to a power supply terminal **H20**. The third magnetoresistive effect element **3P** has a second terminal electrically connected a first terminal of the third magnetoresistive effect element **3Q**. The third magnetoresistive effect element **3Q** has a second terminal electrically connected the reference terminal **L10**. The third output terminal **3T** is electrically connected the connection point between the pair of third magnetoresistive effect elements **3P** and **3Q**.

(43) The fourth magnetoresistive effect element **4P** has a first terminal electrically connected to the reference terminal **L20**. The fourth magnetoresistive effect element **4P** has a second terminal electrically connected a first terminal of the fourth magnetoresistive effect element **4Q**. The fourth magnetoresistive effect element **4Q** has a second terminal electrically connected to the power supply terminal **H20**. The fourth output terminal **4T** is electrically connected to the connection point between the pair of fourth magnetoresistive effect elements **4P** and **4Q**.

(44) The first output terminal **1T**, the second output terminal **2T**, the third output terminal **3T**, and the fourth output terminal **4T** are electrically connected the processing circuit **201**. Note that in FIGS. 1 and 2, only the first output terminal **1T** is connected to the processing circuit **201** for simplification.

(45) In FIGS. 1, 2, 4, 5, and 6, the shape of each magnetoresistive effect element **Mr0** is rectangular when viewed in the Z-axis direction. However, this shape is schematically drawn to show the orientation of the magnetoresistive effect element **Mr0** and does not correspond to the actual shape of the magnetoresistive effect element **Mr0**.

(46) The magnetoresistive effect element **Mr0** has an electrical resistance value that varies according to the magnitude of a magnetic field applied. The magnetic sensor **100** outputs, as a voltage signal, the variation in the electrical resistance value of the magnetoresistive effect element **Mr0**. The magnetoresistive effect element **Mr0** has no sensitivity to a magnetic field in a first direction (direction along the long side in FIG. 1) but has sensitivity to a magnetic field in a second direction (direction along the short side in FIG. 1). The sensitivity of the magnetoresistive effect element **Mr0** is maximum to the magnetic field in the second direction.

(47) The pair of first magnetoresistive effect elements **1P** and **1Q** and the pair of third magnetoresistive effect elements **3P** and **3Q** are arranged to have sensitivity to a magnetic field in a direction along the X-axis. The pair of first magnetoresistive effect elements **1P** and **1Q** and the pair of third magnetoresistive effect elements **3P** and **3Q** cause a resistance variation in the same pattern to the magnetic field aligned with the positive direction of the X-axis and the magnetic field aligned with the negative direction of the X-axis if the two magnetic fields have the same magnitude.

(48) The pair of second magnetoresistive effect elements **2P** and **2Q** and the pair of fourth magnetoresistive effect elements **4P** and **4Q** are disposed to have sensitivity to a magnetic field in a direction along the Y-axis. The pair of second magnetoresistive effect elements **2P** and **2Q** and the

pair of fourth magnetoresistive effect elements **4P** and **4Q** cause a resistance variation in the same pattern to the magnetic field aligned with the positive direction of the Y-axis and the magnetic field aligned with the negative direction of the Y-axis if the two magnetic fields have the same magnitude.

(49) When viewed in the Z-axis direction, the magnetoresistive effect element **Mr0** are arranged as indicated below with respect to the center of the magnetic sensor **100**. That is, the first magnetoresistive effect element **1P** and the third magnetoresistive effect element **3P** are disposed on a positive side of the Y-axis from the center. The first magnetoresistive effect element **1Q** and the third magnetoresistive effect element **3Q** are disposed on a negative side of the Y-axis from the center. The second magnetoresistive effect element **2P** and the fourth magnetoresistive effect element **4P** are disposed on a positive side of the X-axis from the center. The second magnetoresistive effect element **2Q** and the fourth magnetoresistive effect element **4Q** are disposed on a negative of the X-axis from the center.

(50) As described above, the Z coordinate of each of the four magnetic poles **50** shown in FIG. **1** is greater than the Z coordinate of each of the remaining four magnetic poles **50**. That is, of the plurality of magnetic poles **50** of the bias magnet **5**, the four magnetic poles **50** shown in FIG. **1** face the plurality of magnetoresistive effect elements **Mr0** and apply a bias magnetic field to the plurality of magnetoresistive effect elements **Mr0**. In FIG. **1**, the orientation of the bias magnetic field is indicated by an arrow.

(51) To the first magnetoresistive effect element **1P** and the third magnetoresistive effect element **3P**, The bias magnetic field aligned with the positive direction of the X-axis is applied. To the first magnetoresistive effect element **1Q** and the third magnetoresistive effect element **3Q**, the bias magnetic field aligned with the negative direction of the X-axis is applied.

(52) To the second magnetoresistive effect element **2P** and the fourth magnetoresistive effect element **4P**, the bias magnetic field aligned with the positive direction of the Y-axis is applied. To the second magnetoresistive effect element **2Q** and the fourth magnetoresistive effect element **4Q**, the bias magnetic field aligned with the negative direction of the Y-axis is applied.

(53) Thus, the single bias magnet **5** generates the bias magnetic field aligned with the positive direction of the X-axis and the bias magnetic field aligned with the negative direction of the X-axis. Moreover, the single bias magnet **5** generates the bias magnetic field aligned with the positive direction of the Y-axis and the bias magnetic field aligned with the negative direction of the Y-axis.

(54) The magnetoresistive effect element **Mr0** is a giant magnetoresistance (GMR) element. More specifically, the magnetoresistive effect element **Mr0** is a current-in-plane (CIP) type GMR element. As shown in FIG. **7**, the magnetoresistive effect element **Mr0** includes a multilayer stack **90** and an undercoat layer **93**.

(55) The multilayer stack **90** is formed by alternately stacking, one on top of another, a plurality of magnetic layers **91**, each containing NiFeCo as a component, and a plurality of non-magnetic layers **92**, each containing Cu as a component. Such a structure enables providing a high-output magnetoresistive effect element **Mr0**. The number of layers of the multilayer stack **90** may be, for example, either greater than or equal to 10 or greater than or equal to 20. The magnetic layer **91** is a layer of a ferromagnetic material. The magnetic layer **91** is magnetized more easily than the non-magnetic layer **92**. The non-magnetic layer **92** preferably contains Cu only.

(56) As shown in FIG. **7**, the magnetic sensor **100** includes a substrate layer **6**. The substrate layer **6** is included in the wiring layer **W1** (see FIG. **3**). The substrate layer **6** includes a substrate **61** and a glazing layer **62**. The glazing layer **62** is formed on a surface of the substrate **61**. The glazing layer **62** includes, as a material, a glass material such as amorphous glass. The glazing layer **62** is formed by printing glass paste onto the surface of the substrate **61** and firing the glass paste. On a surface of the glazing layer **62**, the magnetoresistive effect element **Mr0** is formed.

(57) The multilayer stack **90** is laid on top of the undercoat layer **93**. More specifically, the undercoat layer **93** is formed on the surface of the glazing layer **62** of the substrate layer **6**, and the

multilayer stack **90** is formed on a surface of the undercoat layer **93**. The undercoat layer **93** contains NiFeCr as a component thereof. Providing the undercoat layer **93** enables providing a high-output magnetoresistive effect element **Mr0**. In addition, providing the undercoat layer **93** allows crystal grains in the magnetic layer **91** to grow sufficiently, thus improving the heat resistance of the magnetoresistive effect element **Mr0**.

(58) The magnetoresistive effect element **Mr0** has no sensitivity in a predetermined direction but has sensitivity isotropically in a plurality of directions intersecting with the predetermined direction.

(59) The bias magnet **5** applies a magnetic field (bias magnetic field) having a field strength of less than or equal to one half field strength of an anisotropic magnetic field of each of the plurality of magnetoresistive effect elements **Mr0** to each of the plurality of (eight) magnetoresistive effect elements **Mr0** including the pair of first magnetoresistive effect elements **1P** and **1Q** and the pair of second magnetoresistive effect elements **2P** and **2Q**. This enables reducing the distortion of an output waveform of each of the plurality of magnetoresistive effect elements **Mr0**.

(60) (6) Protection Film

(61) As shown in FIG. **3**, the first protection film **71** covers the wiring layer **W1**. The bias magnet **5** is mounted on a surface of the first protection film **71**. The second protection film **72** covers the bias magnet **5**.

(62) The first protection film **71** is made of, for example, a resin, a metal oxide such as Al.sub.2O.sub.3 (alumina), or a metal nitride. The second protection film **72** is made of, for example, a resin.

(63) (7) Processing Circuit

(64) The processing circuit **201** (refer to FIG. **1**) includes a computer system including one or more processors and a memory. The functions of the processing circuit **201** are performed by making the processor of the computer system execute a program stored in the memory of the computer system. The program may be stored in the memory. Alternatively, the program may also be downloaded via a telecommunications network such as the Internet or distributed after having been stored in a non-transitory storage medium such as a memory card.

(65) The processing circuit **201** determines, based on the first, second, third, and fourth output signals, the orientation of the magnetic field applied to the magnetic sensor **100**. The first output signal, the second output signal, the third output signal, and the fourth output signal are respectively output from the first output terminal **1T**, the second output terminal **2T**, the third output terminal **3T**, and the fourth output terminal **4T**. In other words, the first output signal, the second output signal, the third output signal, and the fourth output signal are respectively output from the first half-bridge circuit **1**, the second half-bridge circuit **2**, the third half-bridge circuit **3**, and the fourth half-bridge circuit **4**.

(66) Comparing the first half-bridge circuit **1** and the third half-bridge circuit **3** with each other, it can be seen that the two half-bridge circuits **1** and **3** have the same sensitivity direction of the magnetoresistive effect elements **Mr0** and the same orientation of the bias magnetic fields but that the arrangement of the magnetoresistive effect elements **Mr0** on the higher- and lower-potential sides in one of these two half-bridge circuits **1** and **3** is opposite from the arrangement in the other half-bridge circuit **3** and **1** as shown in FIGS. **1** and **5**. Thus, the third output signal has a reverse phase with respect to the first output signal.

(67) Comparing the second half-bridge circuit **2** and the fourth half-bridge circuit **4** with each other, it can be seen that the two half-bridge circuits **2** and **4** have the same sensitivity direction of the magnetoresistive effect elements **Mr0** and the same orientation of the bias magnetic fields but that the arrangement of the magnetoresistive effect elements **Mr0** on the higher- and lower-potential sides in one of these two half-bridge circuits **2** and **4** is opposite from the arrangement in the other half-bridge circuit **4** and **2** as shown in FIGS. **1** and **6**. Thus, the fourth output signal has a reverse phase with respect to the second output signal.

(68) (8) Detection of Magnetic Field Orientation

(69) The magnetic sensor **100** is disposed in the vicinity of the rotor **8**. The plurality of magnetic poles **80** of the rotor **8** form a magnetizing field. As the rotor **8** rotates, the orientation of the magnetic field applied to the magnetic sensor **100** changes. The processing circuit **201** determines, based on an output of the magnetic sensor **100**, the orientation of the magnetic field applied to the magnetic sensor **100**.

(70) Note that supposing that not the rotor **8** but the magnetic sensor **100** rotates with respect to the rotor **8**, the orientation of the magnetic field applied to the magnetic sensor **100** also changes as the magnetic sensor **100** rotates, and the processing circuit **201** may also determine the orientation of the magnetic field. Thus, in the following description, a situation where the rotor **8** is fixed and the magnetic sensor **100** changes its position in the order of L1, L2, L3, L4, and L5 will be described with reference to FIG. 4. The magnetic sensor **100** rotates around the rotor **8**, and therefore, the X-axis and the Y-axis also rotate accordingly.

(71) At each of the positions L1, L2, L3, L4, and L5, the magnetic sensor **100** is located radially outside of the rotor **8**. In this case, the orientation of the magnetic field applied to the magnetic sensor **100** is perpendicular to the direction defined by a rotational axis of the rotor **8**. Thus, the orientation of the magnetic sensor **100** needs to be adjusted to make the Z-axis defined for the magnetic sensor **100** aligned with the direction of the rotational axis of the rotor **8**.

(72) As the magnetic sensor **100** changes its position in the order of L1, L2, L3, L4, and L5 (actually, as the rotor **8** rotates), each of the first, second, third, and fourth output signals changes in either a sinusoidal wave or a cosine wave. FIG. 8 shows a waveform V1 of the first output signal and a waveform V2 of the second output signal. The third output signal is a signal having a reverse phase with respect to the first output signal. The fourth output signal is a signal having a reverse phase with respect to the second output signal. Thus, illustration of the third and fourth output signal is omitted from FIG. 8. In FIG. 8, the plurality of magnetic poles **80** are illustrated as straight ones for the sake of convenience.

(73) When the magnetic sensor **100** is located at the position L3 where the magnetic sensor **100** faces the center of one of N-magnetic poles **80** of the rotor **8**, a magnetic field aligned with the positive direction of the Y-axis is applied to the magnetic sensor **100**. The pair of first magnetoresistive effect elements **1P** and **1Q** and the pair of third magnetoresistive effect elements **3P** and **3Q** detect no magnetic field. The bias magnetic field aligned with the positive direction of the Y-axis is applied to the second magnetoresistive effect element **2P** and the fourth magnetoresistive effect element **4P**, and therefore, the magnetic field of the rotor **8** and the bias magnetic field enhance each other. On the other hand, the bias magnetic field aligned with the negative direction of the Y-axis is applied to the second magnetoresistive effect element **2Q** and the fourth magnetoresistive effect element **4Q**, and therefore, the magnetic field of the rotor **8** and the bias magnetic field attenuate each other.

(74) Therefore, when the magnetic sensor **100** is at the position L3, the second output signal is minimum and the fourth output signal is maximum.

(75) When the magnetic sensor **100** is located at the position L1, L5 where the magnetic sensor **100** faces the center of one of S-magnetic poles **80** of the rotor **8**, the orientation of the magnetic field of the rotor **8** is opposite to the orientation in the case of the magnetic sensor **100** being at the position L3, and therefore, the second output signal is maximum and the fourth output signal is minimum.

(76) When the magnetic sensor **100** is located at the position L2 where the magnetic sensor **100** faces a boundary between an N-magnetic pole **80** and an S-magnetic pole **80** of the rotor **8**, a magnetic field aligned with the positive direction of the X-axis is applied to the magnetic sensor **100**. The pair of second magnetoresistive effect elements **2P** and **2Q** and the pair of fourth magnetoresistive effect elements **4P** and **4Q** detect no magnetic field. The bias magnetic field aligned with the positive direction of the X-axis is applied to the first magnetoresistive effect element **1P** and the third magnetoresistive effect element **3P**, and therefore, the magnetic field of

the rotor **8** and the bias magnetic field enhance each other. On the other hand, the bias magnetic field aligned with the negative direction of the X-axis is applied to the first magnetoresistive effect element **1Q** and the third magnetoresistive effect element **3Q**, and therefore, the magnetic field of the rotor **8** and the bias magnetic field attenuate each other.

(77) Thus, when the magnetic sensor **100** is located at the position **L2**, the first output signal is maximum and the third output signal is minimum.

(78) The rotor **8** has a plurality of boundaries each between an N-magnetic pole **80** and an S-magnetic pole **80** in a circumferential direction. When the magnetic sensor **100** is located at the position **L4** where the magnetic sensor **100** next faces a boundary between an N-magnetic pole **80** and an S-magnetic pole **80** after the position **L2**, the orientation of the magnetic field of the rotor **8** is opposite to the orientation in the case where the magnetic sensor **100** is located at the position **L2**, and therefore, the first output signal is minimum and the third output signal is maximum.

(79) As shown in FIG. **8**, every time a relative rotational angle of the magnetic sensor **100** with respect to the rotor **8** changes by a rotational angle corresponding to double the width of each magnetic pole **80**, the first and second output signals will repeat the same waveform. In other words, the rotational angle corresponding to double the width of each magnetic pole **80** corresponds to one cycle of each of the first and second output signals.

(80) Supposing the first and second output signals are each a sinusoidal wave, the phase difference between the first and second output signals is a rotational angle corresponding to one half the width of each magnetic pole **80**. That is to say, the phase difference is a quarter of one cycle. Therefore, supposing the first output signal is a sinusoidal wave, the second output signal will be a cosine wave with respect to the first output signal.

(81) For example, the processing circuit **201** derives, based on the first and second output signals, a common phase between the first output signal as a sinusoidal wave and the second output signal as a cosine wave. The processing circuit **201** may decide that every time the phase has changed by one cycle, the magnetic sensor **100** (actually, the rotor **8**) have rotated to a rotational angle corresponding to one cycle. In other words, the processing circuit **201** may decide that every time the phase has changed by one cycle, the magnetic sensor **100** (actually, the rotor **8**) have rotated to a rotational angle corresponding to double the width of each magnetic pole **80**. In this manner, the processing circuit **201** may determine how much the magnetic sensor **100** (actually the rotor **8**) has rotated from the rotational angle at the starting point. That is to say, the processing circuit **201** may determine a relative rotational angle.

(82) Also, the phases of the first and second output signals correspond to the orientation of the magnetic field applied to the magnetic sensor **100**. That is to say, the processing circuit **201** may determine the orientation of the magnetic field applied to the magnetic sensor **100**. More specifically, the processing circuit **201** may determine the orientation of the magnetic field applied to the magnetic sensor **100** within a range of 0 degrees to 360 degrees.

(83) In another example, the processing circuit **201** may determine, based on not only the first and second output signals but also the third and fourth output signals, the rotational angle of the magnetic sensor **100** (actually the rotor **8**). Specifically, the processing circuit **201** generates a first differential signal representing the difference between the first output signal and the third output signal. The first differential signal has a waveform, of which the amplitude is double the amplitude of the first output signal. In addition, the processing circuit **201** also generates a second differential signal representing the difference between the second output signal and the fourth output signal. The second differential signal has a waveform, of which the amplitude is double the amplitude of the second output signal. The processing circuit **201** derives, based on the first and second differential signals, a common phase between the first differential signal as a sinusoidal wave and the second differential signal as a cosine wave. The processing circuit **201** may decide that every time the phase has changed by one cycle, the magnetic sensor **100** (actually, the rotor **8**) have rotated to a rotational angle corresponding to one cycle. The amplitude of the first and second

differential signals is double the amplitude of the first and second output signals, thus allowing the orientation of the magnetic field and the rotational angle of the magnetic sensor **100** (actually the rotor **8**) to be determined more accurately.

(84) Optionally, the magnetic detection system **200** may include a sensor (such as an optical sensor or a magnetic sensor) for detecting the starting point of movement (e.g., rotation) of the target of measurement (e.g., the rotor **8** in this case). Every time the target of measurement makes one full rotation, the sensor generates a predetermined output signal and the processing circuit **201** detects the starting point based on the predetermined output signal.

(85) (First Variation)

(86) A first variation of the embodiment will be described below. In the following description, any constituent element of this first variation, having the same function as a counterpart of the embodiment described above, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(87) In the embodiment described above with reference to FIG. **4**, the magnetic sensor **100** is disposed outside of the circumference of the rotor **8**. However, this is only an example and should not be construed as limiting. Alternatively, the magnetic sensor **100** may also be disposed at a position **L41** (refer to FIG. **4**), for example. That is to say, the magnetic sensor **100** may also be disposed at a position where the magnetic sensor **100** faces the rotor **8** in a direction parallel to the rotational axis of the rotor **8**. Even in that case, as the rotor **8** rotates, the magnetic field applied to the magnetic sensor **100** also rotates to allow the magnetic sensor **100** to detect the orientation of the magnetic field.

(88) In that case, however, the orientation of the magnetic sensor **100** needs to be different from that of the embodiment described above. The orientation of the magnetic field applied to the magnetic sensor **100** will be perpendicular to the radial direction of the rotor **8**. Thus, the orientation of the magnetic sensor **100** needs to be adjusted such that the Z-axis defined for the magnetic sensor **100** is aligned with the radial direction of the rotor **8**.

Other Variations of Embodiment

(89) Next, other variations of the embodiment will be enumerated one after another. Note that the variations to be described below may be adopted in combination as appropriate. Optionally, the variations to be described below may also be adopted as appropriate in combination with the first variation described above.

(90) The magnetic sensor **100** does not have to be used to detect the rotational angle of the target of detection. Alternatively, the magnetic sensor **100** may also be used to detect a linear movement of a target of detection.

(91) The target of detection does not have to be the rotor **8**. Optionally, the target of detection and an object that generates magnetism to be detected by the magnetic sensor **100** may be formed separately from each other and then attached to each other.

(92) The magnetic sensor **100** does not have to be used to determine the rotational angle but only needs to be used to detect the orientation of the magnetic field.

(93) The reference terminal **L20** electrically connected to the first magnetoresistive effect element **1P** and the reference terminal **L20** electrically connected to the fourth magnetoresistive effect element **4P** may be separate terminals. Similarly, the reference terminal **L10** and the power supply terminals **H10** and **H20** may be individually disposed for each magnetoresistive effect element **Mr0**.

(94) Optionally, the third half-bridge circuit **3** and the fourth half-bridge circuit **4** may be omitted from the magnetic sensor **100**. In that case, the processing circuit **201** may determine the orientation of the magnetic field applied to the magnetic sensor **100** based on the first and second output signals, instead of the first differential signal representing the difference between the first and third output signals and the second differential signal representing the difference between the second and fourth output signals.

(95) In the magnetic sensor **100**, the first protection film **71** and the second protection film **72** may

be omitted.

(96) The number of the bias magnet **5** is not limited to one, but two or more bias magnets **5** may be provided.

(97) The number of magnetic poles **50** of the bias magnet **5** is not limited to eight, but, for example, two or four magnetic poles **50** may be provided.

(98) (Recapitulation)

(99) The embodiment and its variations described above are specific implementations of the following aspects of the present disclosure.

(100) A magnetic sensor (**100**) of a first aspect includes at least one bias magnet (**5**), a first half-bridge circuit (**1**), a second half-bridge circuit (**2**), and a base member (**73**). The at least one bias magnet (**5**) is configured to generate a bias magnetic field aligned with a positive direction of an X-axis, a bias magnetic field aligned with a negative direction of the X-axis, a bias magnetic field aligned with a positive direction of a Y-axis orthogonal to the X-axis, and a bias magnetic field aligned with a negative direction of the Y-axis. The base member (**73**) holds the at least one bias magnet (**5**), the first half-bridge circuit (**1**), and the second half-bridge circuit (**2**). The first half-bridge circuit (**1**) includes a pair of first magnetoresistive effect elements (**1P**, **1Q**) half-bridge connected to each other and configured to detect a magnetic field aligned with the X-axis and a first output terminal (**1T**) configured to output a first output signal from a connection point between the pair of first magnetoresistive effect elements (**1P**, **1Q**). The second half-bridge circuit (**2**) includes a pair of second magnetoresistive effect elements (**2P**, **2Q**) half-bridge connected to each other and configured to detect a magnetic field aligned with the Y-axis and a second output terminal (**2T**) configured to output a second output signal from a connection point between the pair of second magnetoresistive effect elements (**2P**, **2Q**). The bias magnetic field aligned with the positive direction of the X-axis is applied to one of the pair of first magnetoresistive effect elements (**1P**, **1Q**), and the bias magnetic field aligned with the negative direction of the X-axis is applied to the other of the pair of first magnetoresistive effect elements (**1P**, **1Q**). The bias magnetic field aligned with the positive direction of the Y-axis is applied to one of the pair of second magnetoresistive effect elements (**2P**, **2Q**), and the bias magnetic field aligned with the negative direction of the Y-axis is applied to the other of the pair of second magnetoresistive effect elements (**2P**, **2Q**).

(101) This configuration causes the first output signal delivered as the magnetic field applied to the magnetic sensor (**100**) rotates to have a waveform close to an ideal sinusoidal wave and causes the second output signal to have a waveform close to an ideal cosine wave. This makes it accurate to determine, based on the first output signal and the second output signal, the orientation of the magnetic field applied to the magnetic sensor (**100**).

(102) In a magnetic sensor (**100**) of a second aspect referring to the first aspect, the at least one bias magnet (**5**) includes a single bias magnet (**5**) configured to generate the bias magnetic field aligned with the positive direction of the X-axis and the bias magnetic field aligned with the negative direction of the X-axis.

(103) This configuration, as compared with the case where a bias magnet for generating a bias magnetic field aligned with the positive direction of the X-axis and a bias magnet for generating a bias magnetic field aligned with the negative direction of the X-axis are individually provided, reduces an angle error between the two bias magnetic fields. This enables reducing the distortion of the waveform of the first output signal.

(104) In a magnetic sensor (**100**) of a third aspect referring the second aspect, the at least one bias magnet (**5**) includes a single bias magnet (**5**) configured to generate the bias magnetic field aligned with the positive direction of the X-axis, the bias magnetic field aligned with the negative direction of the X-axis, the bias magnetic field aligned with the positive direction of the Y-axis, and the bias magnetic field aligned with the negative direction of the Y-axis.

(105) This configuration enables, as compared with the case where four bias magnets are provided

in accordance with four bias magnetic fields, reducing an angular error among the four bias magnetic fields. This enables reducing the distortion of the waveforms of the first output signal and the second output signal.

(106) In a magnetic sensor (**100**) of a fourth aspect referring to any one of the first to third aspects, among the pair of first magnetoresistive effect elements (**1P**, **1Q**) or the pair of second magnetoresistive effect elements (**2P**, **2Q**), at least one magnetoresistive effect element (**Mr0**) includes a multilayer stack (**90**) formed by alternately stacking, one on top of another, magnetic layers (**91**) containing NiFeCo as a component and non-magnetic layers (**92**) containing Cu as a component.

(107) This configuration contributes to increasing the output of the magnetoresistive effect element (**Mr0**).

(108) In a magnetic sensor (**100**) of a fifth aspect referring to the fourth aspect, the at least one magnetoresistive effect element (**Mr0**) includes an undercoat layer (**93**) containing NiFeCr as a component and the multilayer stack (**90**) laid on top of the undercoat layer (**93**).

(109) This configuration contributes to increasing the output of the magnetoresistive effect element (**Mr0**).

(110) In a magnetic sensor (**100**) of a sixth aspect referring to any one of the first to fifth aspects, the at least one bias magnet (**5**) is configured to apply, to each of a plurality of magnetoresistive effect elements (**Mr0**) including the pair of first magnetoresistive effect elements (**1P**, **1Q**) and the pair of second magnetoresistive effect elements (**2P**, **2Q**), a magnetic field having a field strength of less than or equal to one half field strength of an anisotropic magnetic field of the each of the plurality of magnetoresistive effect elements (**Mr0**).

(111) This configuration may reduce the distortion of the waveforms of the first output signal and the second output signal.

(112) A magnetic sensor (**100**) of a seventh aspect referring to any one of the first to sixth aspects further includes a third half-bridge circuit (**3**) and a fourth half-bridge circuit (**4**). The third half-bridge circuit (**3**) is configured to output a third output signal having a reverse phase with respect to the first output signal. The fourth half-bridge circuit (**4**) is configured to output a fourth output signal having a reverse phase with respect to the second output signal.

(113) This configuration enables the output to be approximately doubled by deriving a differential output between the first output signal and the third output signal, as compared with the first output signal. Similarly, this configuration enables the output to be approximately doubled by deriving a differential output between the second output signal and the fourth output signal, as compared with the second output signal. This enables the orientation of the magnetic field to be more accurately determined based on the differential output.

(114) Note that the constituent elements according to the second to seventh aspects are not essential constituent elements for the magnetic sensor (**100**) but may be omitted as appropriate.

(115) A magnetic detection system (**200**) of an eighth aspect includes the magnetic sensor (**100**) according to any one of the first to seventh aspects and a processing circuit (**201**). The processing circuit (**201**) is configured to determine, based on at least the first output signal and the second output signal, an orientation of a magnetic field applied to the magnetic sensor (**100**).

(116) This configuration may provide a magnetic detection system (**200**) including a processing circuit (**201**) as an integral part thereof.

REFERENCE SIGNS LIST

(117) **1** First Half-Bridge Circuit **1P**, **1Q** First Magnetoresistive Effect Element **2** Second Half-Bridge Circuit **2P**, **2Q** Second Magnetoresistive Effect Element **3** Third Half-Bridge Circuit **4** Fourth Half-Bridge Circuit **5** Bias Magnet **73** Base Member **90** Multilayer Stack **91** Magnetic Layer **92** Non-Magnetic Layer **93** Undercoat Layer **100** Magnetic Sensor **200** Magnetic Detection System **201** Processing Circuit **Mr0** Magnetoresistive Effect Element **1T** First Output Terminal **2T** Second Output Terminal

Claims

1. A magnetic sensor comprising: at least one bias magnet configured to generate a bias magnetic field aligned with a positive direction of an X-axis, a bias magnetic field aligned with a negative direction of the X-axis, a bias magnetic field aligned with a positive direction of a Y-axis orthogonal to the X-axis, and a bias magnetic field aligned with a negative direction of the Y-axis; a first half-bridge circuit; a second half-bridge circuit; and a base member holding the at least one bias magnet, the first half-bridge circuit, and the second half-bridge circuit, the first half-bridge circuit including a pair of first magnetoresistive effect elements half-bridge connected to each other and configured to detect a magnetic field aligned with the X-axis and a first output terminal configured to output a first output signal from a connection point between the pair of first magnetoresistive effect elements, the second half-bridge circuit including a pair of second magnetoresistive effect elements half-bridge connected to each other and configured to detect a magnetic field aligned with the Y-axis and a second output terminal configured to output a second output signal from a connection point between the pair of second magnetoresistive effect elements, the bias magnetic field aligned with the positive direction of the X-axis being applied to one of the pair of first magnetoresistive effect elements, the bias magnetic field aligned with the negative direction of the X-axis being applied to the other of the pair of first magnetoresistive effect elements, the bias magnetic field aligned with the positive direction of the Y-axis being applied to one of the pair of second magnetoresistive effect elements, the bias magnetic field aligned with the negative direction of the Y-axis being applied to the other of the pair of second magnetoresistive effect elements.
2. The magnetic sensor of claim 1, wherein the at least one bias magnet includes a single bias magnet configured to generate the bias magnetic field aligned with the positive direction of the X-axis and the bias magnetic field aligned with the negative direction of the X-axis.
3. The magnetic sensor of claim 2, wherein the at least one bias magnet includes a single bias magnet configured to generate the bias magnetic field aligned with the positive direction of the X-axis, the bias magnetic field aligned with the negative direction of the X-axis, the bias magnetic field aligned with the positive direction of the Y-axis, and the bias magnetic field aligned with the negative direction of the Y-axis.
4. The magnetic sensor of claim 1, wherein among the pair of first magnetoresistive effect elements or the pair of second magnetoresistive effect elements, at least one magnetoresistive effect element includes a multilayer stack formed by alternately stacking, one on top of another, magnetic layers containing NiFeCo as a component and non-magnetic layers containing Cu as a component.
5. The magnetic sensor of claim 4, wherein the at least one magnetoresistive effect element includes an undercoat layer containing NiFeCr as a component and the multilayer stack laid on top of the undercoat layer.
6. The magnetic sensor of claim 1, wherein the at least one bias magnet is configured to apply, to each of a plurality of magnetoresistive effect elements including the pair of first magnetoresistive effect elements and the pair of second magnetoresistive effect elements, a magnetic field having a field strength of less than or equal to one half field strength of an anisotropic magnetic field of the each of the plurality of magnetoresistive effect elements.
7. The magnetic sensor of claim 1, further comprising: a third half-bridge circuit configured to output a third output signal having a reverse phase with respect to the first output signal; and a fourth half-bridge circuit configured to output a fourth output signal having a reverse phase with respect to the second output signal.
8. A magnetic detection system comprising: the magnetic sensor of claim 1; and a processing

circuit configured to determine, based on at least the first output signal and the second output signal, an orientation of a magnetic field applied to the magnetic sensor.
